

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/335812629>

# Advance designs of progressive addition lenses over conventional

Preprint · September 2019

DOI: 10.13140/RG.2.2.18070.09281

CITATIONS

0

READS

93

1 author:



[Debapriya Mukhopadhyay](#)

Rajeev Gandhi University of Health Sciences Karnataka

5 PUBLICATIONS 41 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Progressive Lens [View project](#)



Optometry Professional Development [View project](#)

# Advance designs of progressive addition lenses over conventional



By  
Debapriya Mukhopadhyay  
M.Optom , First Year,  
B.V.P.U

To  
My teachers  
My college  
My parents  
And  
My sister Shreyasi

| <u>Contents –</u>                                                          | Page  |
|----------------------------------------------------------------------------|-------|
| Introduction to progressive addition lenses –                              |       |
| 1. History of Progressive addition lenses.                                 | 3     |
| 2. What are progressive addition lenses?                                   | 4-5   |
| 3. Basic fundamentals of progressive addition lenses                       | 5-6   |
| 4. What are the needs of progressive lenses?                               | 7     |
| Designs of conventional progressive addition lenses –                      |       |
| 1. What are conventional progressive lenses?                               | 8     |
| 2. Bifocal vs. conventional progressive lenses                             | 8-9   |
| 3. Entry to the progressive addition lens world                            | 10    |
| 4. Variations of conventional progressive lenses                           | 11-20 |
| Advance designs of progressive addition lenses –                           |       |
| 1. What are the advance designs?                                           | 20-21 |
| 2. What are the basic differences between advance design over conventional | 21-22 |
| 3. What are the needs of advance design                                    | 23-24 |
| 4. Advance design PAL comes with...                                        | 25-26 |
| Marketing of advanced design PALs –                                        |       |
| 1. What are the advance design PALs available in market?                   | 26-38 |
| 2. Frequently asked questions                                              | 39-41 |
| 3. Advance design PAL frame selection                                      | 42    |
| 4. Caring of advanced design progressive addition lenses                   | 42-43 |
| 5. Progressive lens Complaints                                             | 43    |
| Marking of advanced design PALs –                                          |       |
| 1. Fitting tips                                                            | 43-44 |
| 2. Ditest                                                                  | 44    |
| 3. Troubleshooting                                                         | 45    |
| Conclusion –                                                               |       |
| Progressive addition lenses for all.                                       | 45    |
| References –                                                               |       |
| The books & web addresses                                                  | 46    |

## History of progressive addition lenses.

The first patent for a PAL was British Patent 15,735, granted to Own Aves with a 1907 priority date. Aves' patent included the progressive lens design and the manufacturing process. However this was unlike modern PALs. It consisted of a conical back surface and a cylindrical front with opposing axes in order to create a power progression. This design was never commercialized.

While there were several intermediate steps (H. Newbold appears to have designed a similar lens to Aves around 1913), there is evidence to suggest that Duke Elder in 1922 developed the worlds first commercially available PAL (Ultrifo) sold by "Gowlland of Montreal". This was based on an arrangement of aspherical surfaces.

The Varilux lens was the first PAL of modern design. It was developed by Bernard Maitenaz, patented in 1953, and introduced by the Société des Lunetiers (that later became part of Essilor) in 1959.

Early progressive lenses were relatively crude designs but modern sophisticated progressive lenses have gained greater patient acceptance and include special designs to cater to many separate types of wearer application: for example lenses may be customized for use with computers, or to offer enlarged near and intermediate view areas. Over the 1980s through today, manufacturers have been able to minimize unwanted aberrations by:

1. Improvements in mathematical modeling of surfaces, allowing greater design control.
2. Extensive wearer trials.
3. Improved manufacturing and lens metrology technology

Today the complex surfaces of a progressive lens can be cut and polished on computer-controlled machines, allowing 'freeform surfacing', as opposed to the earlier casting process.

## What are progressive addition lenses?

Progressive addition lenses (PAL) are corrective lenses used in eyeglasses to correct presbyopia and other disorders of accommodation. They are characterised by a gradient of increasing lens power, added to the wearer's correction for the other refractive errors. The gradient starts at a minimum, or no addition power, at the top of the lens and reaches a maximum addition power, magnification, at the bottom of the lens. The length of the progressive power gradient on the lens surface is usually between 15 and 20 mm with a final addition power between 1.00 to 2.50 dioptres for most wearers. The addition value prescribed depends on the level of presbyopia of the patient and is closely related to age.

### Advantages of progressive addition lenses (Conventional)

- The wearer can adjust the additional lens power required for clear vision at different viewing distances by tilting his or her head to sight through the appropriate part of the vertical progression;
- The lens location of the correct addition power for the viewing distance usually only requires small adjustments to head position, since near vision tasks such as reading are usually low in the visual field and distant objects higher in the visual field.
- Progressive addition lenses avoid the discontinuities (image-jumps) in the visual field created by bifocal and trifocal lenses and are more cosmetically attractive. Since bifocal and related designs are associated with 'old age', proponents have suggested the lack of segments on the lens surface of a progressive lens appears more 'youthful' since lenses associated with younger wearers single vision lenses tend to be free of segments or lines on the surface.

### Disadvantages of progressive addition lenses (Conventional)

Distortion: Progressive lenses suffer the disadvantage of the power progression creating regions of aberration away from the optic axis, yielding poor visual resolution (blur). As the lenses combine a range of powers in a single surface there are also geometric distortions to the visual field, which increase with the addition power.

Fitting: Progressive lenses require careful placement relative to the wearers pupil centre for a distance'-viewing reference position. Incorrect specification of the fitting location can cause problems for the wearer including narrow fields of view, clear vision in one eye only, on-axis blur, and the need to adopt uncomfortable head positions.

Cost: Progressive lenses are sold at a significantly higher price than bifocal and single-vision reading spectacles.

### Basic fundamentals of progressive addition lenses –

In PALs there is a continuous downward transition of lens power from stable distance power in upper portion of the lens through a progressive zone to a stable near vision portion in the lower area of the lens. This progressive zone extends across the width of the lens and connects the distance and near vision portions. The central portion of the progressive zone is called as progressive corridor. There is no visible reading segment and there are no dividing lines, hence there is no image jump. On the both side of progressive corridor there some astigmatism occurs because a progressive addition surface is produced by generating an aspheric curvature.



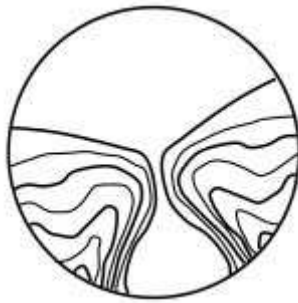
The inherited astigmatism is proportional to the rate of change in curvature. There are two basic approaches to the design of progressive addition lenses.

1. Hard design
2. Soft design

### Hard Design progressive addition lenses –

The hard design is created by having short progressive corridor because the amount of surface astigmatism generated is a function of the rate of progression. These designs concentrate the intense aberrations into small regions adjacent to the progressive corridor and near add area. In this case wearer having large distance and near zones that

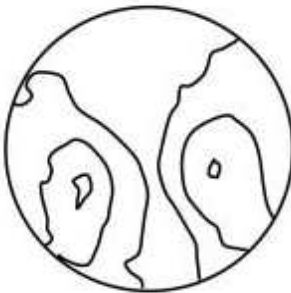
are free of astigmatism. As the near vision area is located closer to the distance centre so the near vision area is more accessible with downward rotation of the eyes. The negative points of the hard design lenses are the areas of high aberrations adjoining the progressive corridor and the near areas produce strongly blurred vision and spatial distortion.



*Hard Design PAL*

Soft design progressive addition lenses –

Soft designs have a long progression corridor that extends into the lateral portion of the distance and near areas of the lens, as if there is no such heavy concentration of aberrations in any one area the intensity of the resulting astigmatism and visual disturbances are reduced. This design provides smaller areas of high quality imagery in the distance and near vision portion of the lens. The soft design imposes a lower location of the near vision add area, which forces greater eye or head movements to use that area.



*Soft Design PAL*



## The needs of progressive lenses –

Both children and adults are likely to benefit from ongoing studies into progressive lenses. In one recent American study, myopia or nearsightedness has been shown to progress more slowly in children who wear glasses with progressive lenses than in those who don't. Until the past few years, children with myopia were prescribed single-vision eyeglass lenses designed to adjust only blurred distance vision. When myopic children of various ages were provided with glasses with progressive lenses, however, their myopia showed a slower rate of development.

Another recent study of progressive lenses indicates that those developed with wavefront technology are regarded as more effective by eyeglass wearers. Wavefront technology uses an instrument known as an aberrometer to measure refractive error in the eye and provides quick and comprehensive information about vision errors. This enables lens prescriptions for progressive lenses to become increasingly accurate, helping to improve the vision of eyeglass wearers whether they have problems with near or distant vision.

Progressive lenses developments have also extended to sunglasses, with a variety of options available in this area. Lens prescriptions can be provided for people who want progressive lenses in their sunglasses, even the wraparound variety of glasses. These glasses can be particularly beneficial for drivers who normally wear glasses and have to make a constant series of vision adjustments when driving, such as glancing from the odometer to the road ahead.

Progressive lenses are among the most popular forms of treatment for people affected by presbyopia. An age-related condition, presbyopia is a form of farsightedness that makes it hard to focus clearly on near objects. Glasses with progressive lenses make it easier for people to cope with presbyopia, with the strength of the lens prescription depending on the level of vision correction needed by the person wearing the glasses.

Signs that a person may be suffering from the onset of presbyopia and would therefore benefit from progressive lenses usually start in middle age. A common symptom that glasses may be needed is a blurring of vision when looking at close-up objects. This makes it hard to perform such actions as reading, especially small print, or to sew or to focus on the information on a computer screen. Reading material often has to be held at arm's length and, if still trying to function without eyeglass assistance, the person may experience headaches and eyestrain, as well as a feeling of tiredness. These problems could be helped successfully by professionally prescribed progressive lenses.

## What are conventional progressive lenses?

In a conventional progressive lens, an uninterrupted series of horizontal curves links distance vision intermediate vision, and near vision with no visible separation. Lens power increases smoothly from the distance vision area at the top of the lens, through an intermediate vision area in the middle, to the near vision area at the bottom of the lens. The term conventional comes from the usual design of lenses which are mostly use by the wearer from that time when the progressive lenses come. Basically these conventional lenses are hard design.

## Conventional Progressive Lens vs. Bifocal

### Advantages

- o No sudden “image jump” from distance to near.
- o An infinite number of focal points to view objects at different distances.
- o No visible line where the bifocal power begins.
- o More natural vision for near viewing.
- o A variety of lens designs to fit virtually any application.

### Disadvantages

- o Requires a short period of adaptation to progressive lens for first time wearers.
- o Costs more than a regular bifocal.

Progressive lenses, or “no-line” bifocals, are increasing in popularity because of the advantages they offer to wearers of regular “lined” bifocals, along with the many lens designs now available. In addition, the advancements in progressive lens technology have helped reduce their cost over time.

From a safety standpoint, the main reason we recommend that a company cover the cost of progressive lenses for their employees is that many people already wear them in their “street” glasses, and adjusting back to a line bifocal for their daily safety wear can be very difficult, and possibly dangerous. As an individual adjusts back to a line bifocal, their ability to see near objects clearly may be inhibited. This can create a safety hazard, and may also interfere with the quality of their work.

Most companies have found that the additional cost of providing progressive lenses to their employees is outweighed by the many advantages, including increased safety and employee satisfaction.

While progressive lenses are preferred by most people, some are unable to adapt to wearing these lenses comfortably. This is mainly due to slight distortions at the

peripheral areas of the bifocal, caused by the “blending” of the bifocal. For most people, this is not a concern, but for those that cannot adjust to this, a regular line bifocal may be the best option.

## Comparison of Bifocal and Progressive Addition Lenses on Aviator Target Detection Performance

*Markovits AS, Reddix MD, O'Connell SR, Collyer PD. Comparison of bifocal and progressive addition lenses on aviator target detection performance. Aviat. Space Environ. Med. 1995; 66:303-8.*

The objective of this project was to determine if the type of presbyopic correction worn by aviators, conventional bifocal versus progressive addition lenses (PAL's), differentially affects aviator visual search performance. Experienced aviators, most with tactical fighter aircraft experience, searched for high-contrast targets under simulated dawn/dust (mesopic) lighting conditions ( $\sim 3.0$  cd/m<sup>2</sup>) while wearing either a standard bifocal (ST-25) or PAL spectacle correction. Latency of locating high-contrast targets under these viewing conditions was differentially affected by the type of presbyopic correction used. Specifically, compared to a standard bifocal (ST-25), a PAL correction (Varilux Infinity®) significantly lowered the time needed to locate static targets at a cockpit instrument viewing distance (83 cm). Accuracy of target location responses was not affected by the type of correction used. In addition, 7 months post-experiment, 7 of the 12 participants (58%) indicated that they used their PAL correction exclusively when flying the T-39 Sabre Liner. Three subjects (25%) used their PAL correction intermittently (primarily at night) when flying and two subjects preferred not to use the PAL's. These results suggest that relative to bifocals, speed of responding to static targets at intermediate viewing distances may be improved by wearing PAL's, and that subjects were able to adapt to PAL lenses quickly in a laboratory setting, using them later in a functional aviation environment.

Aviators are required to view critical information at a minimum of three accommodation distances (approach plate, cockpit instruments, and infinity). Bifocals, however, have only 2 focal lengths; 20 feet (infinity) and, usually, 40 cm, a fairly standard reading distance. Trifocals on the other hand, although offering a correction for three viewing distances, necessarily have a relatively small intermediate segment (vertically), preventing a full view of the cockpit instruments (because trifocals are almost never used by military aviators, they were not included for study here). In addition, some pilots are uncomfortable with the head movements that are needed to accommodate changes in focal length when wearing multifocal lenses. These conditions have created a reluctance in pilots to wear presbyopic corrective eyewear, and potential flight hazards associated with inefficient or difficult vision.

Few objective studies have examined the performance of subjects wearing a PAL correction for presbyopia. Previous research comparing PAL wearers to non presbyopic or multifocal controls (i.e., bifocal wearers) in simple target detection and reading tasks suggests that, for near vision ( $\gg 40$  cm):-

1. Compared to pre presbyope controls, presbyopes wearing PAL's have a normal range of eye movements during simple target detection.
2. Peripheral target detection time is no different between practiced PAL and trifocal wearers.
3. Adaptation to PAL's may involve a combination of anticipatory head movements, adjustments in saccadic gain, and acquisition of visual cues from a slightly blurred retinal image. These aspects of eye-head coordination may be fundamentally different for PAL and non-PAL wearers.
4. Reading rate, reading comprehension, and eye movements associated with reading have been observed to be the same for bifocal and PAL wearers.
5. Normal head movement is induced for targets  $\leq \pm 7^\circ$  from central fixation for PAL wearers versus  $\pm 8.5^\circ$  for bifocal wearers.

## The entry to the progressive world

It was as far back as the beginning of the 20th century that the first steps were taken to produce lenses with a continuous gradual change in power from the distance to the reading prescription. One of the first attempts employed a parabolic concave back surface with an increasing radius of curvature from the centre of the lens, creating a corresponding decrease in back surface power and, hence, an increasing total lens power. It meant, therefore, that the central portion was used for distance vision and the periphery for progressively nearer vision. The first surface employing the now familiar umbilical meridian with its progressive corridor was designed very soon after. Difficulties in production, the absence of computers to help generate appropriate surfaces and the presence of unacceptable aberrations, all probably accounted for the lack of any commercial success with these early PALs. Most designers since that time have been striving to produce a lens surface in which the radii of curvature between the distance and near portions of the front convex surface could become progressively smaller by equal amounts, horizontally and vertically. At the same time, the surface astigmatism at the lens periphery would need to be reduced to a level that could be tolerated by the patient during normal viewing conditions. It should also be remembered that for optimal binocular vision, the images produced by both lenses must be formed on corresponding retinal positions and have similar optical properties. In all directions of gaze, vertical prismatic effect, as well as power and astigmatism, must be approximately equal.

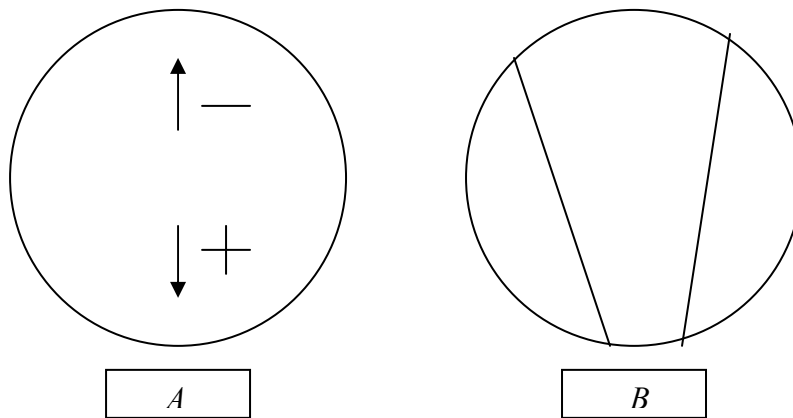
## Variations of progressive lenses –

**The Omnifocal lens**, first commercially successful progressive addition lens introduced in the US. This was invented by David Volk and manufactured by Robinson-Houchin. The front surface of the lens was aspherical on this surface the radius of curvature in the vertical meridian gradually decreased from the top to the bottom of the lens but the horizontal meridian radius of curvature remains same.

The power of the cylinder of the horizontal meridian was  $+0.75\text{ D}$ , greater than the power of the desired addition. That is, for a  $+1.50\text{ D}$ , add; the cylinder on the front surface had a power of  $+2.25\text{ D}$  on the front surface of the lens. The spherical power increasing gradually from top to bottom with a constant amount of astigmatism, created a plus cylinder axis  $90^\circ$

The distance and near vision centers of the omnifocal lens are separated by a vertical distance of 25 mm. the power of the vertical meridian varies over the entire extent of the lens the total amount of plus power decreases from the distance optical centre upward but increases from the distance optical centre downward.

The increase in plus power as the line of sight moves downward through the lens is nonlinear, which results in a maximum addition of  $+1.75\text{ D}$  for a lowering of the line of sight by 25 mm but only one-fourth of this amount is added to the plus power created by lowering the line of sight 12.5mm in this lens progression occurs over the entire surface of the lens this lens was the epitome of a soft design.

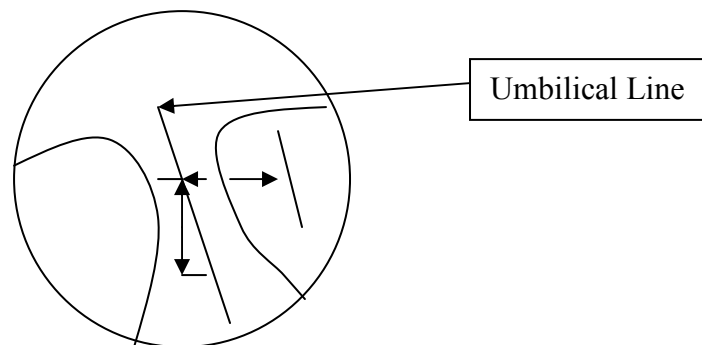


*A) The Omnifocal lens design, the plus power in the distance vision portion decreases from the distance optical centre upward. B) The usable area of the lens is wide at the top and gradually narrows toward the bottom.*

**The Varilux lens** developed by Bernard Maitenaz and the original Varilux lens was introduced by Essel Optical in France in 1959. The original Varilux lens now called the Varilux 1, differed from the Omnifocal in that the upper half of the lens had no progression in power the progression was confined to a 12 mm deep zone in the centre of the lens, throughout which the power increased in a linear fashion. Below the progressive corridor was a zone of maximum addition having a constant power and a width of about 22 mm.

The astigmatism free progressive corridor of the original Varilux lens was approximately 5 mm wide. The usable width of the progressive corridor decreased with increasing power of the add. The distance portion of Varilux lenses has a conventional spherical surface. At the beginning of the progressive zone, without any visible line of demarcation the radius of curvature begins to shorten and uniformly decreases over a distance of 12 mm until the full power of addition is reached. Varilux progressive surface is confined to a 12 mm deep zone rather than extending from top to bottom, the *Varilux lens is considered a hard design*.

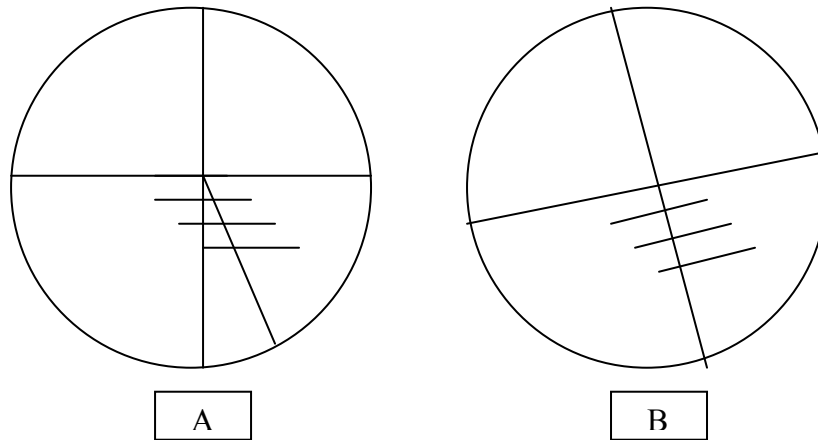
The umbilical line (the line of symmetry at the centre of the progressive corridor) Has some property like for small distances around each point on this line, the vertical and horizontal radii of curvature are essentially equal. For any point along the line of symmetry quasispherical surface properties exist. With the determination of the curvature along the vertical meridian and with the requirement that the line of symmetry from an umbilical line, the progressive line and it's image properties are largely predetermined.



*The amount of astigmatism at a lateral distance from the umbilical line is equal to twice the change in power for the vertical distance.*

Varilux lenses are made for each eye separately with the line of symmetry inclined nasally toward the bottom of the lens, which conforms to the convergence of the visual axes for any point in the progressive zone. The inclination provides a segment inset of 2.5 mm from the distance optical centre.

Other progressive addition lenses achieve segment inset by rotating the segment nasally. On the other hand laboratory finishing to the prescribe distance power, whether spherical or sphero-cylindrical, is accomplished on the back surface in the same manner for any front surface multifocal lens. In the United States the original varilux lens was available only in glass.



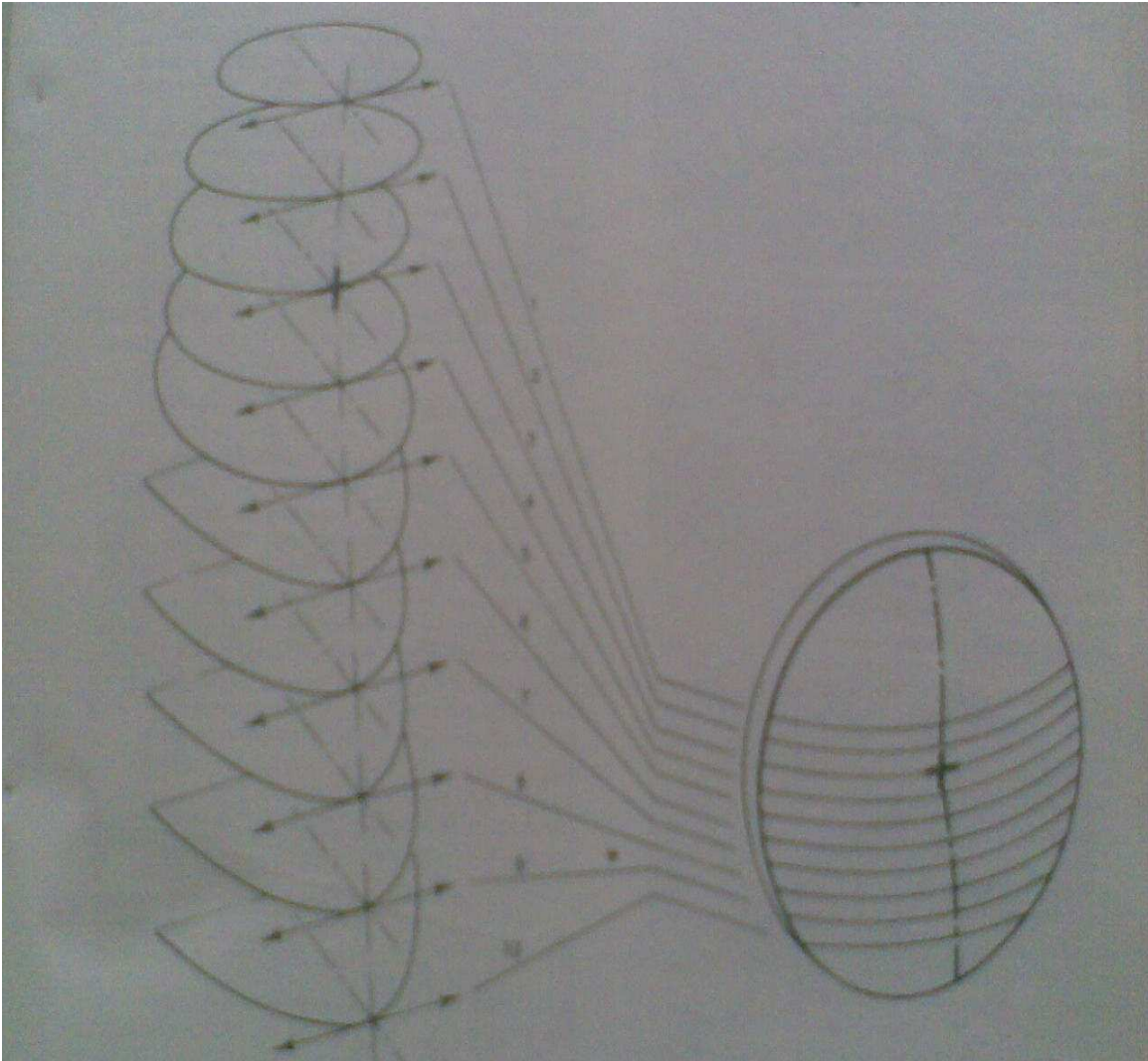
A) With the varilux lens, the line of symmetry is inclined nasally toward the bottom of the lens so equal horizontal eye movements provide uniform power changes.

B) Segment inset is provided by nasal-ward rotation, equal horizontal eye movements fail to provide uniform power changes.

The Varilux 2 lens, a lens in which the entire front surface not just the progressive zone, was of the aspheric design. The Varilux 2 lenses are a family of conic sections is first an oblate ellipse, gradually becomes a circle at the optical centre of the lens, below which it gradually changes to increasingly flatter prolate ellipses, then to a parabola and finally becomes increasingly flatter hyperbolas.

The design reduced the intensity of surface astigmatism and distortion at the periphery of the lens and having a softer design, compared with that of the original varilux lenses. Varilux 2 lenses not only reduced the amount of unwanted astigmatism in the peripheral portion but controlled the axis of the unwanted astigmatism in the peripheral area of the progressive zone. These lenses controlled the axis of unwanted astigmatism in such a way that the rectangularity of peripherally viewed objects was preserved, which thus reduced the rocking and swaying and the disturbing deformation of the image. Now this varilux 2 is called as varilux plus and represents varilux's general purpose, Mono-design, aspheric based, progressive addition lens.

## The Varilux Lens Design.



Starting from the top of the lens as an oblate ellipse it continues as oblate ellipses(2,3) and gradually becomes a circle (4) in the central area then varying prolate ellipses(5,6) and then successively parabolas and hyperbolas(7,8,9,10) in the near vision area.



**The American optical progressive lenses** first introduced the Ultravue lens, a CR-39 plastic lens, this lens have a segment area width of 25mm, which later named as Ultravue25. After Ultravue25 came the lens Ultravue28 which had a segment area of 28mm. The Ultravue differs from Varilux lenses by having a well defined distance-vision portion, free of surface astigmatism, and a relatively wide astigmatism free segment area but at a expense of higher rate of progression in the 10-12 mm progressive corridor and the segment area. This lens is a hard design lens.

The second difference between Ultravue and Varilux lenses is that the progressive corridor in Ultravue lens blank lies vertically rather than being angled nasally toward the bottom of the lens.

In the year of 1982 American Optical introduced the Truvision lenses which replaced the Ultravue. It has a wide segment area and less astigmatism at periphery of the progressive corridor. Later comes Truvision Omni, this lens is described by manufacturers as a soft design with wide intermediate and near vision zones.

**Younger 10/30 Lenses** which was introduced by Younger Optical a PAL in CR-39 plastic lenses. The designation 10/30 used because the lens has a 10 mm deep progressive corridor and a 30 mm wide useable segment area. This lens optimizes the hard PAL design. In this lens image distortion is eliminated both nasally and temporally for a short distance bellow the mid horizontal meridian of the lens. The usable width depends on the base curve, the power of the addition, and the segment height.

The Younger 10/30 is different from the Varilux lens and Ultravue lens because in Younger 10/30 lens the progression is nonlinear as with the Omnifocal lens. The first one third of the power change occurs in the upper 5 mm of the progressive corridor. Remaining 2/3 occurs in the lower 5 mm. It means the amount of unwanted astigmatism outside the progressive corridor increases progressively in the downward direction.

**Cosmetic Parabolic Sphere PALs (CPS)** introduced by Younger Optics. These lenses made up of CR-39 plastic. The basic difference is Younger 10/30 is a hard design lens and CPS lens is a soft design. The CPS lens has an unabated distance area. A 12.5 mm. deep progressive corridor of reasonable width, and a small reading area. The orientation of vertical and horizontal lines of peripherally viewed images is fairly well controlled.

**The Silor Super No-Line Lens** introduced by Silor Optical, which had marketed the original Varilux lens since 1969. This lens is made by Benoit Berthiot of France. This is an improved version of original No-Line PAL; it has a progressive corridor of 12 mm deep. The segment area is about 25mm wide, and this is also a hard design. Surface astigmatism and distortion are present in the peripheral portions of the progressive corridor; the lens is designed to control the axes of unwanted astigmatism to prevent

rotation of vertical and horizontal lines of peripherally viewed objects. Same as the Varilux lens the Super No-Line lens is made with the line of symmetry inclined nasally.

**The Unison Lens** first introduced by Univis Lens Company. A hard design lens, having completely spherical distance area. This lens has a 12 mm deep but relatively narrow corridor and a relatively wide segment area.

**Sola VIP and XL** introduced a CR-39 plastic PAL called the VIP lens. This lens is a soft design lens, having a large distance vision area and a relatively large near vision area with the unwanted astigmatism distributed in the peripheral portion of the lens. Orientation of the vertical and horizontal lines of peripheral object has been well controlled. Sola Optical has introduced a lens called the XL, a soft lens design intended for early presbyopes. Sola introduced VIP Gold and XL Gold lenses which both made in clear Spectralite plastic and Spectralite Transitions. Spectralite having refractive index of 1.537 and Abbe no.47.0, which is the highest Abbe no. of the high index plastics.

### **Other Progressive Addition Lenses**

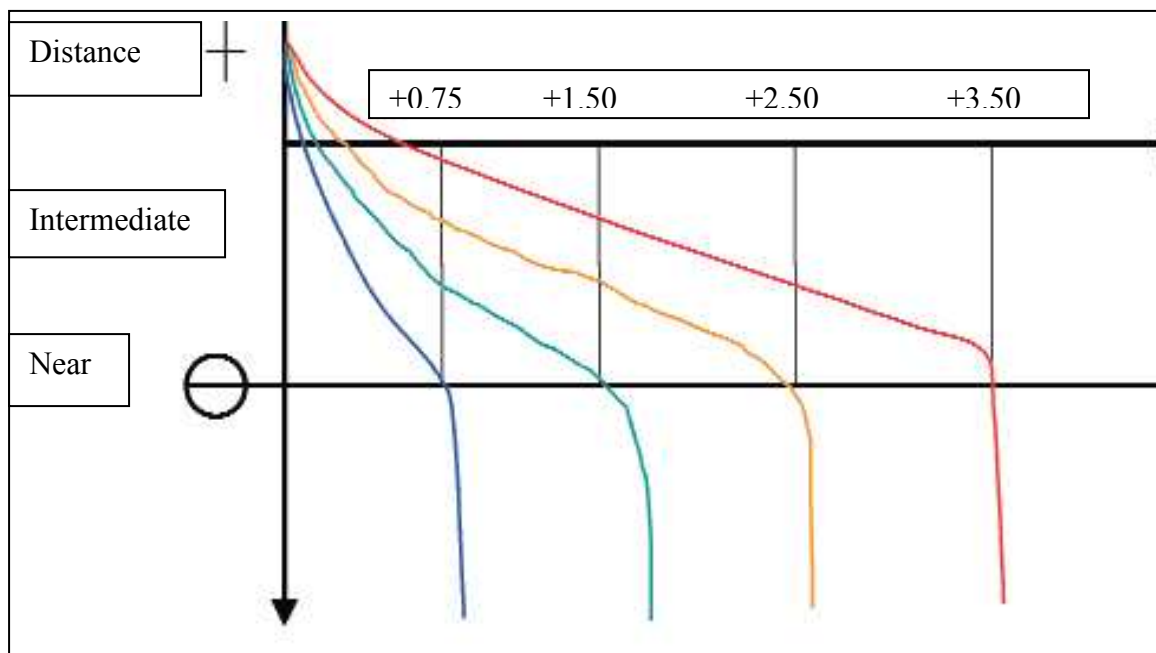
Coburn Optical Industries introduced a lens **Progressiv R**, a lens designed by Rodenstock in Germany. The lens is a hard design lens, having a 12 mm deep corridor and a relatively narrow segment area. To achieve 2 mm segment inset the lens must be rotated 8°. Rodenstock also manufacture the **progressiv S** which is a hard design lens having aspheric surfaces. Using integrated design field-designed aspheric the latest lens is **Progressiv life**.

Seiko Optical Products introduced two progressive addition lenses **P-2**, **P-3**. the P-2 intended for general wear. Is a soft design and reduces the magnitude of astigmatism, but increases the area containing the astigmatic error. The P-3 is recommended for active patients. These both lenses are replaced by **P-6**, a plastic lens described by the manufacturer. P-6 is having linear power progression for active individual.

PolarLite developed a plastic polarized progressive addition lens named Progressive M, which later called as polarlite progressive. This lenses can be tinted with gray or brown, polarlite non-Polaroid progressive lenses are called high time progressive, which having the refractive index of 1.56. this lenses having mild progression leading to easy adaptation.

**Multi design Progressive addition lenses** Until the mid 1980s, all PAL types maintained either hard or soft design features throughout their range of reading additions. An early presbyope who may benefit more from a soft lens design, would be required to change lens types if they were to gain the benefits that a harder lens would provide, as their presbyopia progressed. Switching from a soft to hard design can require considerable adaptation and as the change is generally accompanied by an increase in reading addition,

this will produce even more increased peripheral astigmatism, which will add to the difficulty of adaptation. The dispensing optician is also faced with the difficulty of deciding upon both a suitable alternative lens and the point at which the change would be beneficial. In 1988, Essilor introduced a PAL that used a different design for each reading addition, in an attempt to match the increasing onset of presbyopia. Low reading additions were combined with a soft design, which became harder as the add power increased. The reading area remained almost constant throughout the range. The principle behind this multidesign approach, incorporating the best features of hard and soft lenses, was to ensure visual comfort and ease of adaptation at each stage of presbyopia<sup>3</sup>. This lens was known as the Varilux Multi-Design (VMD) and although not produced anymore, its multi-design concept has been retained with later additions to the Essilor range.



Progression profiles of a multi design progressive addition lens

## Special Purpose Progressive Addition Lenses

COSMOLIT P a lens produced by Rodenstock, is needed for those who having excessive near and intermediate work. The progressive zone is 30 mm long, the power change over that zone is 1.50 D. the power changes 0.05 D per mm.

The fitting of these lenses with the eyes in the primary position of gaze, the fitting cross is dropped 0.5mm down from the papillary centre for every 1° of pantoscopic tilt.

GRADAL RD a lens introduced by Zeiss is a progressive addition lens that is solely for intermediate and near vision activities. Available in both CR-39 and glass of refractive index 1.604 an abbe no. of 41.7, and a specific gravity of 2.67. the lens have a 30 mm long progressive corridor length used in the, compared with the 15 mm corridor length used in the Zeiss Gradal HS general purpose progressive addition lenses.

OVERVIEW a lens made by Varilux, a modified progressive lens designed for those who perform close work above the eye level. The upper part of this lens carries a 41 mm round bifocal segment. The distance between the central fitting cross and the lower edge of the segment is 9 mm. the addition power is always 0.50 less than the engraved progressive addition power.

The fitting of this lens the fitting cross should be exactly at the centre of the pupil with the eyes in primary gaze. The sufficient height of the frame is required.

READABLES are progressive addition lenses that extend the range of clear vision for near and intermediate vision tasks. This lens is manufactured by Varilux these lenses are specially for emmetropic or nearly emmetropic presbyopes.

The power changes 1.50D over a wide corridor in the middle 28mm of the lens. The prescribed near vision power is prescribed at the optical centre. Above the optical centre the power decreases 0.25D every 4mm and stabilizes 12mm above the optical centre at 0.75D less plus than the power of the optical centre.

TECHNICA lens is a progressive addition lens introduced by American Optical. The lens is designed to increase the size of the intermediate lens power area. It is prescribed as a same manner like regular progressive lenses.

The fitting center is located at the center of the pupil. At the fitting center the power is 50% of the near addition power. The intermediate viewing zone is centered 7mm below the fitting center and the power is 80% of the power of the near vision addition. the distance portion of the lens is small and located near the top of the lens but the lens has a very wide usable near vision area.

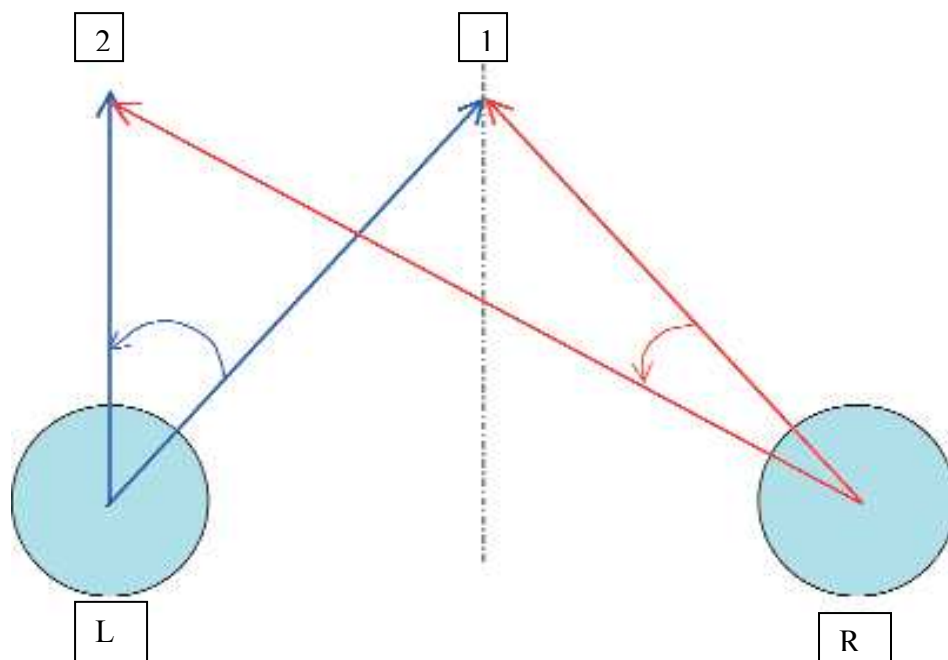
SMARTSEG manufactured by Sola, which has a 30 mm wide straight top bifocal zone with a progressive zone.

The advantages of these lenses

- It extends the intermediate range beyond of a bifocal lens.
- It creates less image jump
- The top of the segment is less noticeable because it is thinner.

### The Asymmetrical Design.

The Varilux® 2, introduced in 1972, was an example of a second generation PAL providing relatively wide fields of view for all distances. It introduced a new concept of 'Horizontal optical moderation'<sup>1</sup> achieved through an asymmetrical design. If both eyes converging to a near point (1) on the midline. As both eyes rotate to fix on point (2), the right eye can be seen to have rotated nasally a lesser amount than the temporal rotation for the left eye. With the new design incorporating a rate of change of surface astigmatism that was greater in the nasal than the temporal regions of the lens, the example given here shows that both eyes will encounter similar amounts of astigmatism at each corresponding point as they look from side to side<sup>2</sup>. Binocular vision was now optimized, with each eye now experiencing equal amounts of surface astigmatism. The lenses, not spherical but a descending succession of conic sections of ellipses, circles, parabolas and hyperbolas, were now produced for right and left eyes with the corresponding near zones already inset.



**The symmetrical design** did mean that lenses could be used for either the right or left eye. Although this suited manufacturer's stock holding requirements, it was not so advantageous for the patient's visual correction. With a symmetrical design, the lens is required to be 'swung' upward nasally in order to try to align the reading zone with the eyes' near visual point when converging to read. With astigmatic prescriptions, one was faced with the issue of compensating the cylinder axis for this lens rotation. This lens, and others designed around this time, did not produce a very high adaptation success rate so it was not surprising that dispensing opticians were reluctant to recommend this as an alternative to bifocal or trifocal lenses.

### **What are the advance designs of progressive addition lenses?**

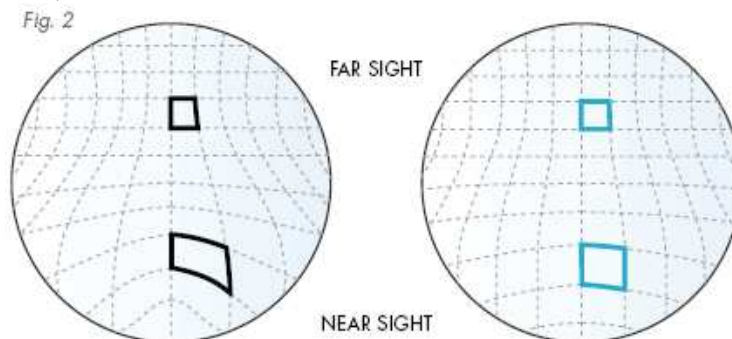
PALs (also known as multifocals or progressives) are spectacle lenses that correct the 3 main areas of vision, namely distance, intermediate and near/reading. Traditional type lenses are very basic designs as the progression is built into the front surface, and the person's distance prescription is ground onto the back surface. For low prescriptions with right and left eyes with similar powers and low astigmatism (cyl power), then this lens is adequate.

New advanced designs are customised to every type of prescription as the progression is created with the person's prescription onto the back surface. This results in vision which has less distortions, less swim effect and consistently wider distance, intermediate and reading zones.

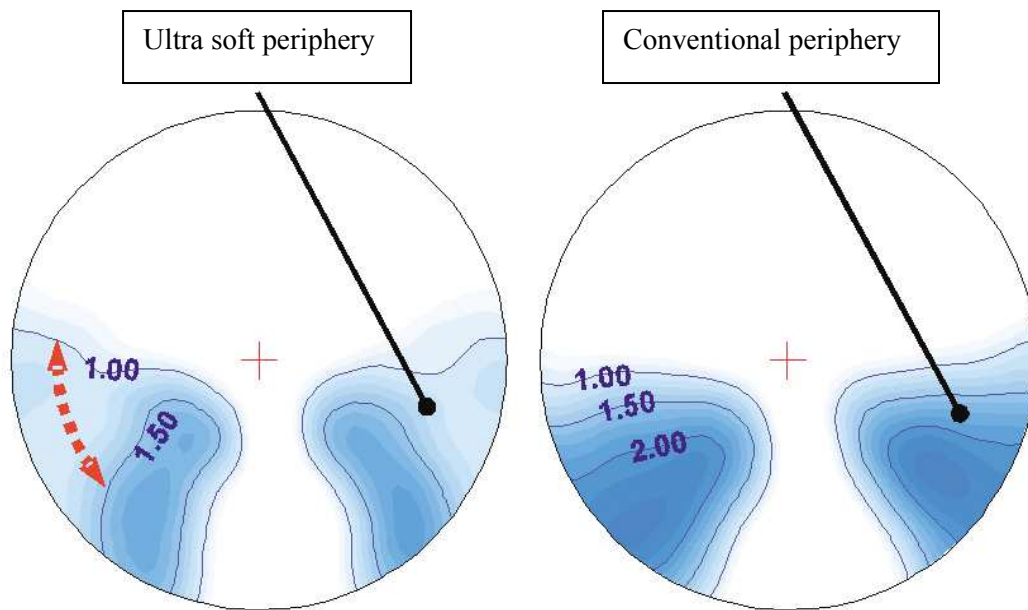
This digitally surfaced (free form) design is ideal for mid- high prescriptions, different right and left powers, reading adds greater than +1.50, and cyls over 1.00D.

Short Corridor multifocal lenses are designed to allow today's multifocal wearers to choose some of the latest small fashion frames. These lenses give excellent distance and reading vision and are available in standard or digitally surfaced designs.

Short corridor multifocals are not recommended for those who use computers for extended periods. Instead use enhanced reading lenses as they offer larger reading and intermediate views, without distance vision.



## Most commonly viewed pictures of differentiations



**Back surface progressive lens**



**Conventional progressive lens**

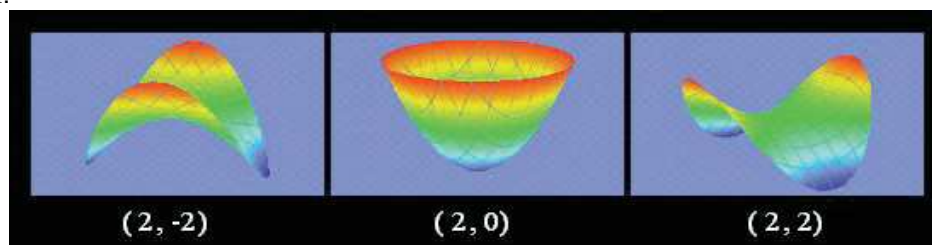
## W.A.V. E. Technology™ Improves Lens Optics

Wave front technology is the most advanced technology used in progressive lenses now a days. Wavefront technology has allowed Essilor researchers to better measure the ability of various progressive lens designs to transfer distortion-free images. Specifically, Varilux Physio lenses are the result of a patented design and manufacturing process called W.A.V. E. Technology: Wavefront Advanced Vision Enhancement™. The process involves identifying the lower and higher order aberrations induced by a lens of a given design, prescription, and material and then fine-tuning the design to eliminate or reduce those aberrations that affect visual acuity and quality of vision. These sophisticated lens designs are precisely reproduced in molds created by digital surfacing technology that is accurate to 0.1 micron. (Varilux Physio 360° also uses digital surfacing to shape the back surface of the lens.) The result is an optimized wavefront produced by the lens—and received by the eye.

### What is wavefront technology?

A wavefront is a sophisticated mathematical concept that describes how a beam of light (the wavefront) is affected when it passes through an optical system. Wavefronts are measured by devices called aberrometers. In the eye, for example, aberrometers measure the wavefront of a beam of light reflected from a point on the fovea. If all the photons from that point were to arrive back at the aberrometer's reference surface simultaneously, the wavefront would be perfectly flat.

If some get to the reference surface a little sooner (due to traveling a shorter path) or a little later, (due to taking a longer path) an irregular wavefront is produced. Any deviation in the wavefront from perfectly flat indicates aberration in the visual system. Scientists know from working with lens optics in astronomy and photography that the flatter the wavefront the more perfect the optical system and the better the modulation transfer function. (Modulation transfer is a term that describes how completely an image is preserved when it passes through a lenticular system, such as the eye or a progressive lens.) In the ideal case, images that pass through a spectacle lens will be undistorted so that patients not only achieve 20/20 acuity but optimal sharpness and contrast sensitivity as well.



A Wavefront map showing the Zernike second-order terms, sphere and cylinder, from phoropter information.



## What are the needs of advance design?

**Quality of Vision** The term “quality of vision” describes the combined quantitative (ie, 20/20 acuity) and qualitative factors that create the optimal visual experience. At present, we lack a universally accepted way to define and measure quality of vision, but its importance to patients is unquestionable. The lessons learned from refractive surgery are now bearing fruit in the realm of spectacle correction for presbyopes. With the advent of Wavefront-optimized progressive lenses, practitioners can now take advantage of Wavefront science to provide presbyopes with an improved visual experience.

**Contrast Sensitivity** Life, unlike acuity charts, does not take place under controlled lighting conditions. Nor does it feature only high-contrast views (ie, sharp-edged black letters printed on a white background). The visual challenges patients encounter each day (and night) are varied and complex; and good contrast sensitivity is an essential component of seeing well across a wide range of conditions. Contrast sensitivity is the optical system’s ability to detect detail, depth, and dimension by quite literally discerning shades of difference between lighter and darker areas such as may exist between objects and their surroundings. Patients who can distinguish fine gradations without difficulty describe their vision as being “sharp” or “crisp.” A number of factors determine a person’s sensitivity to contrast. Age is one—and a reason why helping presbyopes achieve the best possible optical correction is so important. Some factors, such as neural processing, are not remediable by optical correction. However, a key component of the presbyope’s optical system—the progressive lens—is very much within the practice’s control.

**Correcting Aberrations**, Aberrations in the spectacle lens can limit quality of vision even when a patient can read the 20/20 line. In other words, not all 20/20 vision is created equal, and presbyopes’ spectacles play an important role in determining how they perceive and rate their quality of vision. The implications for dispensers are obvious: the higher the patient’s perceived quality of vision, the happier the patient; and the happier the patient, the more successful the practice will ultimately be. Refractive technologies have traditionally corrected only gross aberrations, for example, sphere and cylinder. The advent of Wavefront technology has given clinicians a more sophisticated picture of how an optical medium, such as a progressive lens, acts upon the light that passes through it and how this in turn impacts the quality of the image that is received at the retina.

## Wavefront-optimized Progressive Performance

**Sharpened Distance Acuity** Higher Order Aberrations, particularly coma, are often present in the distance field of progressive lenses. This creates visual distortion and limits the sharpness of focus that can be achieved with conventional progressive designs,

particularly as the eye moves to the lens periphery. By controlling coma in the distance portion of the lens, advanced Wavefront optimized progressive addition lens provides three times less higher order aberration at distance for sharper vision across a widened field.

**Wider Intermediate Vision** As the add power of the lens increases, so does induced astigmatism. This distortion is managed by locating it on either side of the progressive channel. Wavefront optimized progressive addition lens design reduces the impact of this astigmatism by decreasing its amplitude and directing its axis to a vertical position, which studies show is more easily “ignored” by the human visual system. The result is sharper, more natural vision in an intermediate zone that is also up to 30% wider than is available in a traditional progressive lens.

**Extended near Vision** Subtle variations in sphere power (called “defocus” in Wavefront parlance) that occur along the length of the near zone can cause eye strain. Wavefront optimized progressive addition lens design addresses this issue by better controlling the power distribution and reducing the maximum defocus by 10%.<sup>1</sup> This enhances wearers’ viewing comfort by extending the vertical reach of the reading zone and making it more uniform in power.

**Enhanced Contrast sensitivity** When the pupil widens, as it does for distance viewing and in low-light conditions, Higher Order Aberrations have a proportionally greater impact on vision quality. This manifests as decreased contrast sensitivity, night vision symptoms, etc. By decreasing the HOAs created by the eyeglass wearer’s lenses, this technology used lenses optimizes vision, particularly in low-light conditions. In fact, studies show that Wavefront technology used progressive lens increases contrast sensitivity by up to 30% compared to conventional lenses.



## **The accessory of advance design PAL's**

### **Lens Materials**

**Plastic Lenses:** Plastic lenses provide added safety over glass lenses as well as being lighter and slightly thicker. Plastic lenses fog less than glass lenses in humid conditions or cold winter sports such as skiing. Plastic lenses are also better than glass in protecting your eyes from harmful rays that may cause cataracts.

**Glass Lenses:** Glass lenses are stable and offer precise optics, but are fragile and must be tempered to improve their strength. Glass lenses are more resistant to scratching but are 50% heavier than plastic lenses.

**High index plastic lenses:** High Index Plastics allow the manufacturing of lenses which are thinner and lighter than standard plastic lenses. These lenses look best, and vision is enhanced when antireflective coating is applied.

**Polycarbonate Lenses:** Polycarbonate lenses are high index plastic lenses that are virtually unbreakable. They are ideal for children or anyone in a hazardous job or high impact sporting environment. Lighter lenses give more comfort, especially with high prescriptions. The lenses look best, and enhance vision, when anti-reflective coating is applied.

### **Lens Tints & Coatings**

**Anti-Reflective Coating :** Anti-reflective (AR) coating improves night vision by reducing unwanted reflections and glare and are effective at reducing eye fatigue for computer users. AR coating is a must for high prescriptions and high index lenses. AR coating also helps enhance overall appearance by removing all distracting reflections.

**Scratch Resistant Coatings:** Light weight plastic lenses scratch more easily than glass lenses so scratch guard coating helps protect your investment by reducing normal scratching.

**UV Protection Coatings:** Special UV Coating is available for plastic lenses and completely blocks UV light from your eyes and protects them from the harmful UV rays that can cause cataracts. Polycarbonate and high index plastic lenses include built in UV protection at no additional cost.

**Tinted Lens Coatings:** Photochromic lenses lighted indoors and darken in the direct sun. They are now available in glass, polycarbonate and high index lenses.

**Polarized Lenses:** These are the exciting sunglasses that eliminate reflective glare. They are especially appropriate for drivers, fisherman, hunters and all types of outdoor activities.

### **Market availability of progressive addition lenses**

There are several types and brands of progressive lenses are available some of the most important products are as folloing.

#### **Varilux**

##### **General**

##### **1. Physio** Front & front and back design

##### **The needs of wearer**

Ask anyone what they want from their spectacle lenses and the usual answer will be “clear vision”. The challenge for lens designers is to translate this apparently-simple wish into the physical and optical properties of the lens. Vision quality begins (but doesn’t end) with visual acuity. To that end, the impact of power error and astigmatism has been the subject of the detailed studies which are essential to the development of effective lenses. On their own though they are not enough, as other aberrations like coma (a higher-order aberration) can affect image quality. To optimise vision, the full contrast sensitivity function also needs to be considered, as it can be affected by aberrations of all types. The development of Varilux® Physio™ has concentrated on the management of higherorder aberrations.

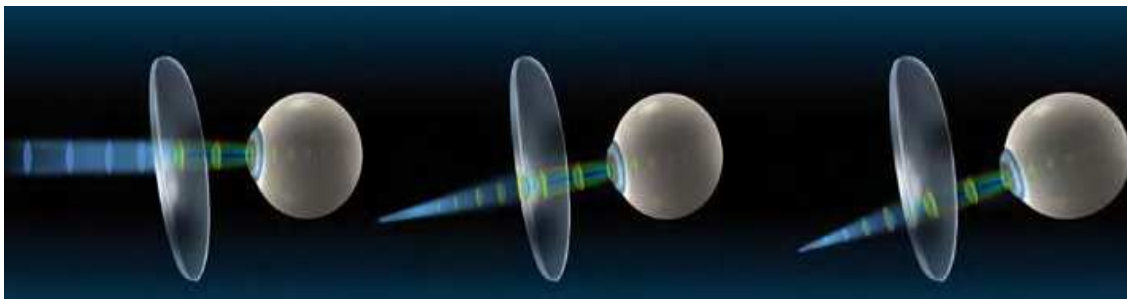
##### **Wavefronts and Aberrations**

Anyone wanting to study higher-order aberrations must begin by thinking of light as travelling as a wave. Wavefronts are virtual surfaces where all points are vibrating in phase. So for a stone dropped into a still pond for example, the concentric circular waves are wavefronts centred on the point of impact. In the case of theoretical optical physics, spherical wavefronts are produced by a point source of light or are focused on a single point. But in the real world, light that has passed through a lens does not focus perfectly and therefore wavefronts exiting the rear surface of a lens are irregular. It is aberrations in the lens that cause the irregularity and the total amount of aberration is the difference between the ideal and actual wavefront. The overall aberration can be broken down into a variety of elementary aberrations which include ‘lower order’ aberrations like power error and astigmatism which in reality have the greatest impact. More complete higher order aberrations include coma, which causes images to flare into a comet shape. Higher order aberrations reduce contrast and image quality. Their effect on vision is

undeniable and increases in proportion with increases in pupil size which is particularly noticeable when the pupil is fully dilated, as in distance viewing and night vision

### Wavefront Management Systems™

The Wavefront Management System™ is the application of wavefront technology to ophthalmic lenses and has led directly to the development of Varilux® Physio™. For each direction of gaze, the Wavefront from the object focused upon is shaped by the ophthalmic lens before entering the eye through its pupil. A study of this Wavefront affords a complete description of the optical properties of the lens for that gaze direction. Breaking this down into its elementary components allows lens designers to put the useful function of the wearer's refraction to one side and concentrate on the remaining elements – the aberrations. In this way, the Wavefront Management System™ is used to calculate the wavefronts for every direction of gaze, allowing the designer to minimise and control aberrations, according to the wearer's individual physiological needs.



*The wavefront is shaped by the lens before passing through the eye's pupil for each position of gaze*

### Visual Needs and Wavefront Management™

#### **Distance Vision**

Varifocal wearers have high expectations of distance vision, both in terms of visual acuity and width of clear vision field. Visual quality is adversely affected by all aberrations – not only power error and astigmatism but also higher order aberrations like coma. These higher order aberrations are to be found in the distance vision zone of varifocals. Use of the Wavefront Management™ system in the development of the Varilux® Physio™ design has helped achieve perfect control of coma meaning that this higher order aberration is fully minimised in a wide sector of the distance vision surrounding the fitting cross. Again, this theoretical benefit was confirmed in extensive wearer testing.

#### **Intermediate Vision**

All varifocals have unwanted residual astigmatism on each side of the meridian in intermediate vision. This astigmatism increases with the addition and the solution has always been to reduce the local amplitude as much as possible. The problem is though that the eye cannot detect local values, instead perceiving the resultant astigmatism across

the entire pupil. The axis of this unwanted astigmatism is also important as the eye naturally prefers to focus on vertical lines to make a line of text easier to read. This is naturally much easier if the axis is positioned vertically. Wavefront Management™ has allowed developers to reduce the amplitude of the resultant astigmatism on the pupil and direct its axis to the vertical position. This gives the wearer a perceived wider field of intermediate vision, an improvement that was verified by wearer testing.

### **Near Vision**

For near vision, a person's posture will change regularly, depending on the task they are performing, their environment and even the time of day. Just as importantly, posture varies from person to person, so if a varifocal is to be acceptable to wearers in every situation, its near vision area must be as wide and high as possible.

## **2. Varilux Comfort Vs. AO Compact**

**Abstract** Over the past several years, small frame styles have increased in popularity and resulted in more patients being fit in PAL's with fitting heights below the traditional 22 to 24 mm minimum recommended height. The American Optical Compact is a new PAL design that is being marketed as a lens that can be used in small B measurement frames, allowing fitting heights at 17mm.

To determine wearer acceptance of PAL's in smaller frames, a clinical study was conducted to compare patient preference and to evaluate lens performance between Varilux Comfort and AO compact when used in small frames with short B. measurement. Clinical findings indicated clear overall preference of Varilux Comfort to AO Compact in every criterion is measured, including near vision.

**Study design** The design of this study was to fit 20 patients in a small frame that would result in a fitting height of 17mm (+/- 1mm). Twenty patients compared the AO Compact lens with Varilux Comfort lens.

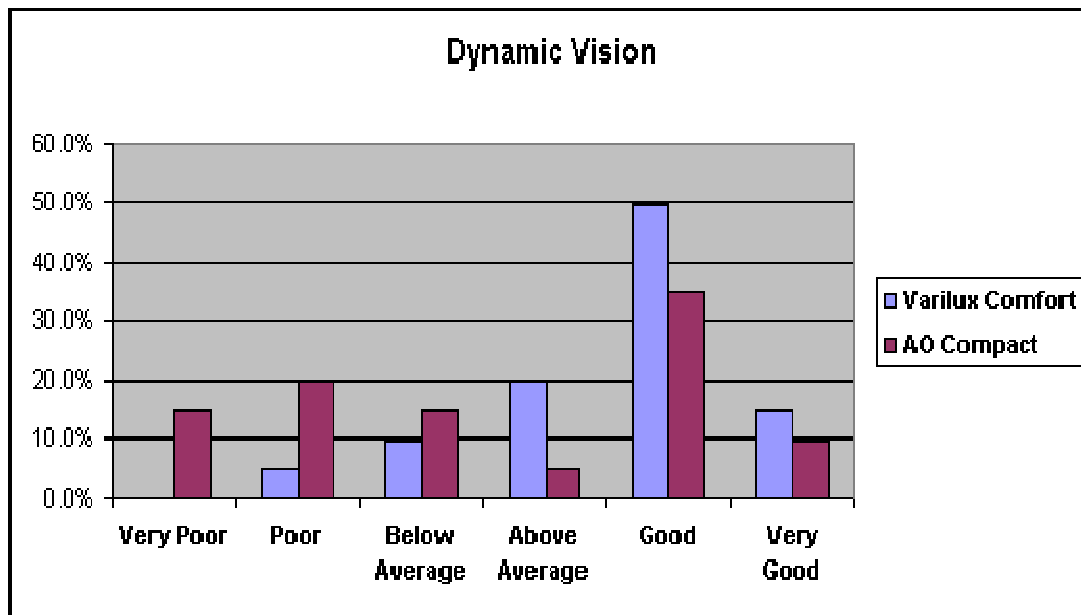
Each patient selected a frame that would allow 17mm fitting height. To avoid order bias, brands were rotated so half of the patients would wear Varilux Comfort first and half would wear AO Compact first. Therefore, each patient was assigned to one of two groups that determined which lens was worn first. The first lens brand was worn for two weeks and then removed from the frame. The second lens was inserted into the same frame, having the same Rx and fitting height as the first, and worn for two weeks.

Each patient filled out four survey questionnaires. The first questionnaire was given prior to wearing either test lens, and was directed at the patient's current glasses. The second questionnaire was given after wearing the test lens for two weeks and was directed at the first test lens. The third questionnaire was given after wearing the second test lens for two weeks and was directed at the second test lens. A final questionnaire was given immediately after the third questionnaire and was directed at the patient's preference of the two test lenses.

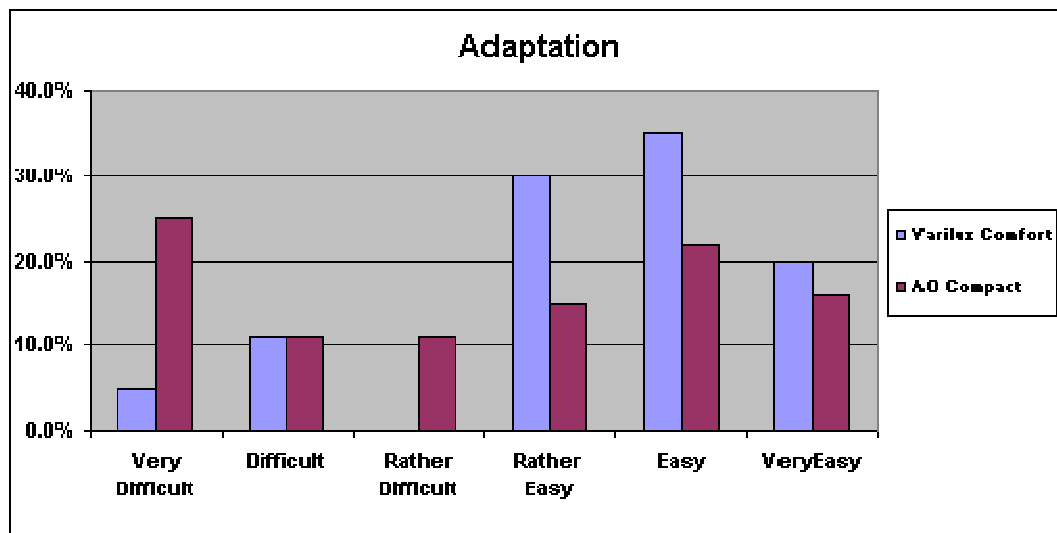
Ten patients wore Varilux Comfort lens first and the AO Compact lens second. Ten patients wore the AO Compact first and the Varilux Comfort second. A list of each

patient's sex, age, occupation, previous Rx, brand fitting height of the test lenses, add power of the test lenses and preferred test lens was tabulated.

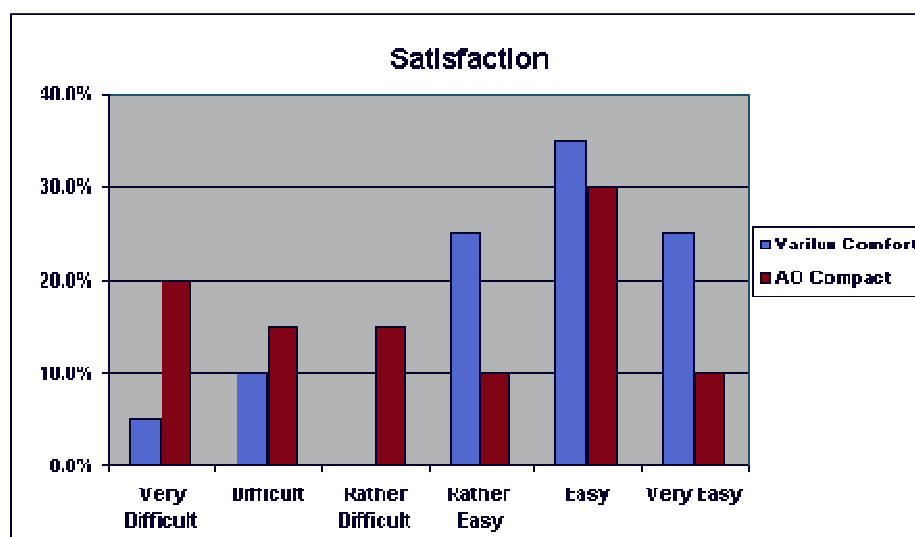
**Dynamic vision** A significant difference in dynamic vision, or vision while moving, was also identified in the patient responses. This is less of a surprise due to the softer peripheral design of Varilux Comfort compared with AO Compact. Eighty five percent of respondents rated dynamic vision with Varilux Comfort as above average to very good compared with only half of respondents with AO Compact. Half of the patients rated dynamic vision with AO Compact as below average to very poor.



**Adaptation** Eighty five percent of patients adapted rather easily to very easily to Varilux Comfort. Fifteen percent of patients found it difficult to very difficult to adapt to Varilux Comfort. Thirty five percent of patients found it difficult to very difficult to adapt to AO Compact; half of the AO Compact patients thought it was rather easy to very easy to adapt to the lens.

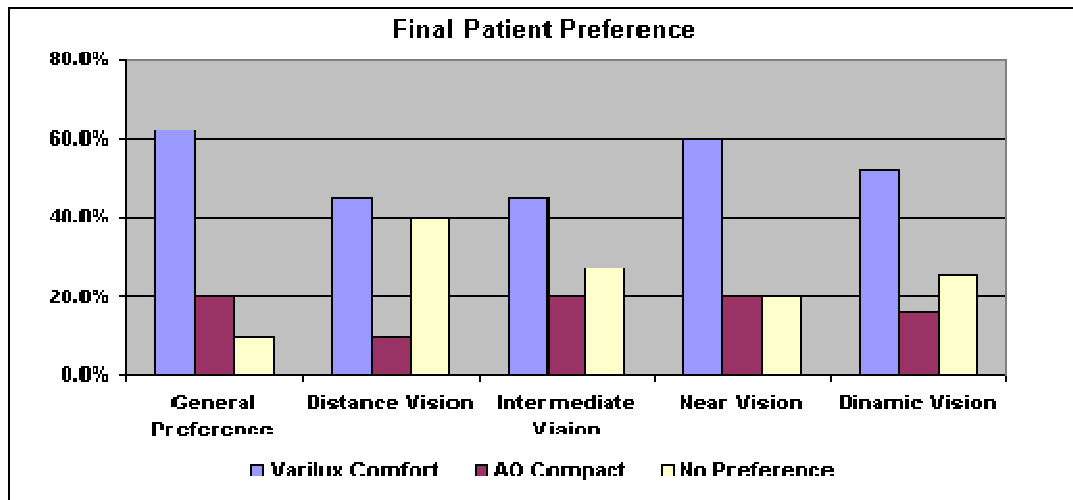


**Satisfaction** Eighty five percent of patients were rather satisfied with Varilux Comfort® compared with half of respondents for AO Compact™. While only 15% of Varilux Comfort patients were dissatisfied to very dissatisfied, more than twice as many patients were dissatisfied to very dissatisfied with AO Compact.





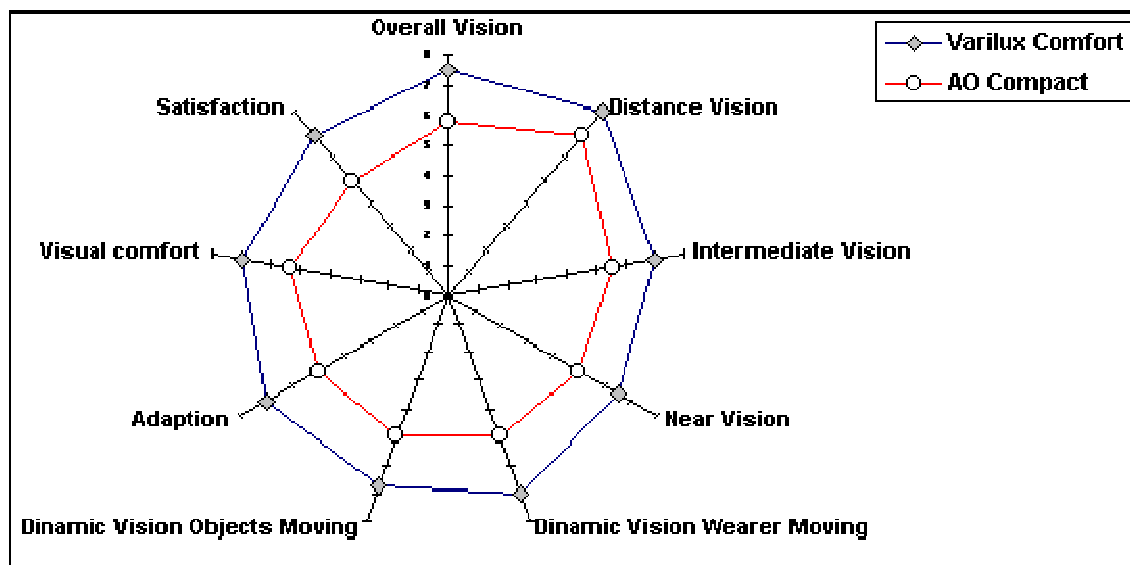
**Preference: Varilux Comfort vs. AO Compact** The average results of the final questionnaire comparing Varilux Comfort to AO Compact indicated a clear overall preference for Varilux Comfort over AO Compact. In fact, 65% of patients preferred Varilux Comfort to AO Compact.



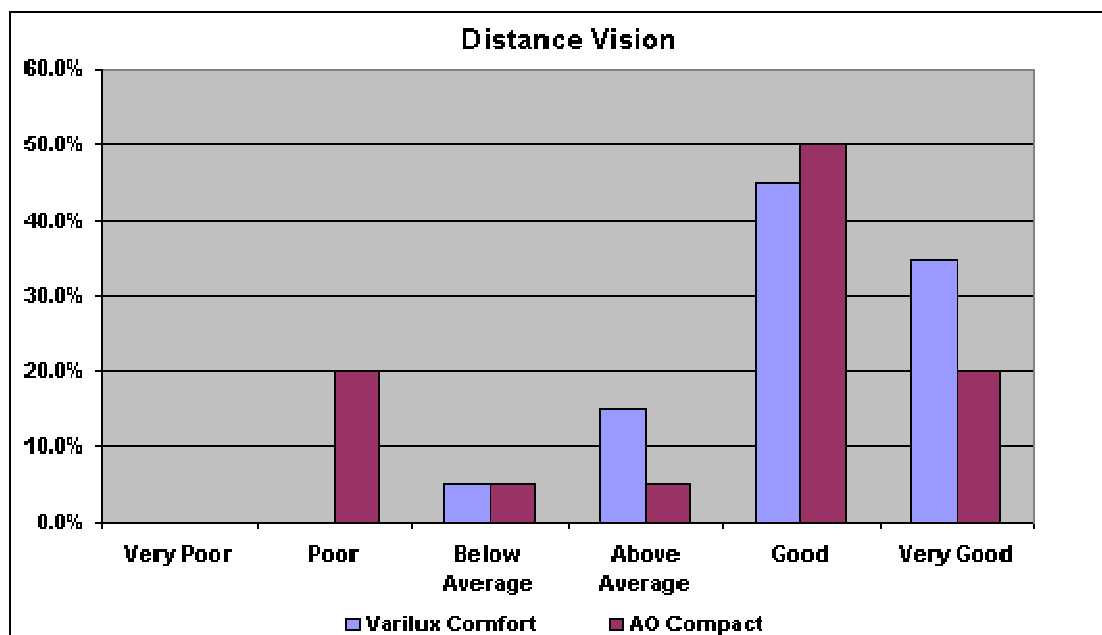
**Patient Profile** The mean patient age was 54 years old with the youngest patient being 44 and the oldest being 67. Seventy five percent of the patients were female. Patients were randomly selected for the clinical test and as a result, previous lenses included single vision reading, bifocals, trifocals, and PAL's. Add powers range from 1.50 diopters to 2.75 diopters.

**Key Findings** Patients were asked to rate their lenses on a variety of criteria, including adaptation, satisfaction, distance, intermediate and near vision. Varilux Comfort® rated higher than AO Compact™ in all criteria tested. Near vision, which is the primary design emphasis of AO Compact, was also rated higher for Varilux Comfort than for AO Compact. Mean satisfaction rating for Varilux Comfort was 7.0 compared with 4.9 for AO Compact.

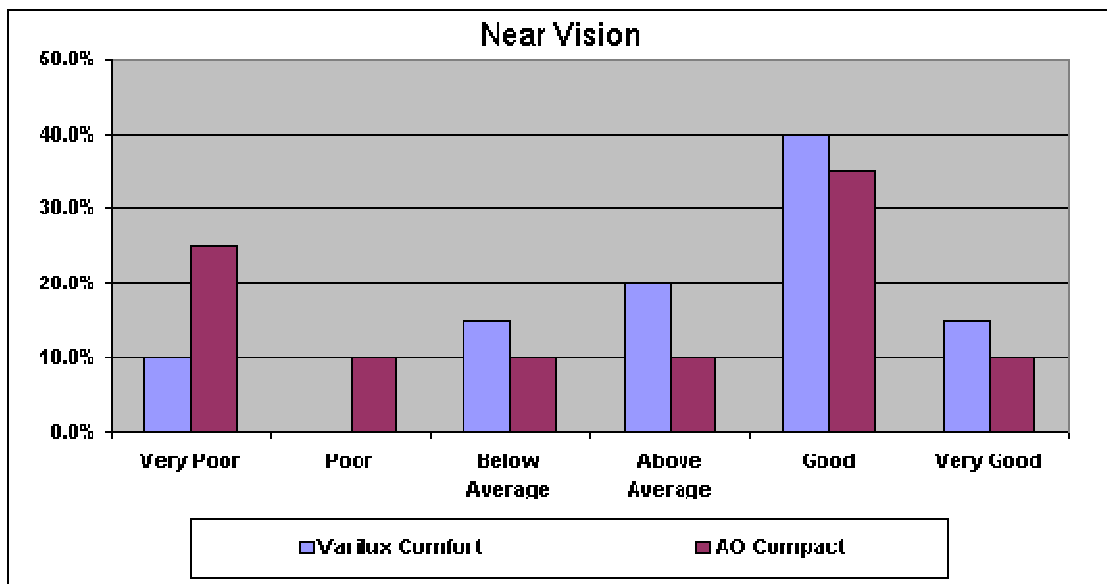
The following radar graph illustrates the average ratings for Varilux Comfort and AO Compact for each of the primary questions. The radar graph utilizes a rating scale of 1 to 10 being the highest. For instance, the mean rating for near vision is 6.5 for Varilux comfort and 5.0 for AO Compact.



**Distance Vision**—Ninety-five percent of patients rated distance vision with Varilux Comfort as above average. Seventy-five percent of patients rated AO Compact as above average with 25% rating it as below average to poor.



**Near Vision** A wider dispersion between the two lenses was identified in near vision. Three quarters of patients rated near vision with Varilux Comfort as being above average to very good with one quarter of patients rating it below average to very poor. In comparison, only slightly more than half of the patients rated near vision with AO Compact as above average to very good. Forty-five percent of patients rated near vision with AO Compact as being below average to very poor. This is somewhat surprising since it was the primary marketing claim for the AO Compact. In the final questionnaire, when patients were asked to rate which lens they preferred for near vision, 60% chose Varilux Comfort, 20 % chose AO Compact and 20% had no preference.



**Conclusion** This study compared the use of Varilux Comfort® and AO Compact™ in a small B size frame with a 17mm fitting height. It showed that although a PAL can be used in a small frame with less than a recommended 22mm fitting height, there are limitations in ease of use and lens performance.

The AO Compact was not preferred over Varilux Comfort and did not perform nearly as well in all areas studied. Patients more often reported significant discomfort when wearing the AO Compact. Overall visual comfort and patient satisfaction were lacking in the AO Compact.

Patients clearly preferred the Varilux Comfort when fit in a small B size frame over the AO Compact. Comfort and ease of use with Varilux Comfort was superior to AO Compact, while the near reading area was functional and as good as AO Compact.

Although a small frame can be used when prescribing a PAL, one would anticipate higher patient ratings if a deeper frame were used that maximizes the near-vision zone.

### **3. Panamic**

The Varilux Panamic design balances central, peripheral, and binocular vision resulting in clear fields of vision no matter where the wearer looks through the lens. The design provides excellent vision correction and wide vision fields in a thin, lightweight lens.

Varilux® Panamic® is the most appropriate choice for patients who are concerned their prescription will call for thick lenses, or require polarized progressive lenses, or are content with their current Varilux® Panamic® lenses and are unwilling to change.

Specialized

#### **1. Ellipse**

Varilux® Ellipse® revolutionized short corridor lens design by providing maximum visual space for more natural vision. Designed specifically for small frames, Varilux® Ellipse® adapts to a small frame wearer's natural visual behavior.

#### **New Varilux® Ellipse® 360° offers an even greater level of performance**

The 360° Digital Surfacing™ process optimizes VARILUX lenses to reduce distortion and maximize fields of vision for each unique prescription. 360° Digital Surfacing uses a Point-by-Point Prescription Mapping™ process to evaluate the interaction of the front side design and the patients' prescription and creates a unique backside design for that prescription. This unique backside design is then applied using 360° Digital Surfacing™ producing an optimal lens for the patient.

Varilux® Ellipse® and Varilux® Ellipse® 360° offer:

- Full scale distance vision with a 140° of clear distance vision
- Easy eye movement to reach near vision quickly
- More comfortable progressive vision in a small frame

Varilux® Ellipse® 360° offers:

Even wider fields of vision with less distortion

The innovative Varilux® Ellipse® design made even more effective for complex prescriptions

#### **2. Liberty**

Varilux® Liberty® is the perfect lens for those on a budget. An ideal entry into the Varilux® range, Varilux® Liberty® lenses feature some of the latest developments in varifocal technology.

Varilux® Liberty® has three main focal areas that are smoothly blended into one for

natural vision at every distance - near far and in between. Because of the design, wearers will also find they adapt more easily to Varilux® Liberty® lenses than to other budget varifocals.

Personalized

### 1. **Ipseo**

#### **The choice of excellence**

Do wearers normally turn their eyes, their head, or both? To what degree? Because each wearer is unique, Essilor has designed Varilux® Ipseo®, the 1st progressive lens adapted to individual visual behavior.

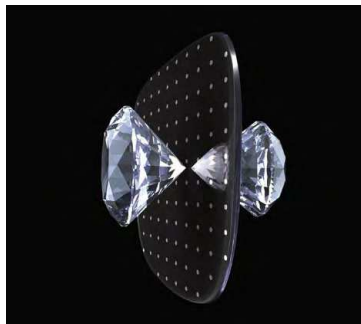
#### **Properties**

Varilux Ipseo lenses are unique, offering unequalled quality of vision. They are the ultimate solution for wearers who are seeking a customized solution for their eyes. The way in which each individual observes his or her surroundings is unique: we all have our own particular visual strategy, an ingenious mix of eye and head movements. The visual strategy of each individual customer can be determined without difficulty by means of a customized diagnosis performed by an eyecare professional. Each Varilux Ipseo lens is then designed individually to match the particular visual strategy of the wearer for whom it is intended.

Thanks to Varilux Ipseo, the wearer can once again enjoy truly natural vision  
Varilux® Ipseo® corrects presbyopia.  
Varilux® Ipseo® is designed for those who want the best for their eyes.

#### **Technology**

The technology behind Varilux Ipseo is based on two major innovations:  
the Wave Front Management System™, a system for calculating the progressive surface of the lens which results in a significant reduction of aberrations  
Point By Point Twinning™ which then allows ultra-precise digital manufacture (to 1/10th of a micron) of the surface calculated in this way.



## Smartpro Progressive Lens availability:-

### Smart Pro Available In:

|                                                |
|------------------------------------------------|
| 1.5 Smart Pro                                  |
| 1.5 Smart Pro Transitions Gray IV              |
| 1.5 Smart Pro Transitions Brown IV             |
| 1.5 Smart Pro Nupolar Gray 1                   |
| 1.5 Smart Pro Nupolar Gray 3                   |
| 1.5 Smart Pro Nupolar Brown                    |
| 1.5 Smart Pro Nupolar Green 1.5                |
| 1.5 Smart Pro Drivewear                        |
| 1.53 Smart Pro Trilogy                         |
| 1.53 Smart Pro Trilogy Transitions V           |
| 1.56 Smart Pro Mid Index                       |
| 1.58 Smart Pro Polycarbonate                   |
| 1.58 Smart Pro Polycarbonate Transitions V     |
| 1.58 Smart Pro Polycarbonate Nupolar Gray      |
| 1.58 Smart Pro Polycarbonate Nupolar Brown     |
| 1.58 Smart Pro Polycarbonate Nupolar Green 1.5 |
| 1.61 Smart Pro High Index                      |
| 1.61 Smart Pro Transitions V                   |
| 1.67 Smart Pro High Index                      |
| 1.67 Smart Pro Transitions V                   |
| 1.67 Smart Pro Polarized Gray                  |
| 1.74 Smart Pro Super High Index                |

Available in minimum fitting heights of 14mm, 17mm, and 19mm

## Kodak Progressive addition lens availability

### Kodak Concise™ Progressive Lenses

**Kodak** Concise is an enhanced varifocal lens designed specifically for modern shallow frames. The compact design of the **Kodak** Concise Lens means that you are not limited to larger frame styles as with some other varifocal designs and still have great vision at all distances.

With a traditional varifocal lens, a minimum depth of frame is required to ensure that you can fit both the distance area and the reading area into the frame. The **Kodak** Concise Lens is designed to fit into a shallow frame but still give you maximum reading area.

Nearly all **Kodak** Concise Lenses are supplied with RLX Plus®, the world's most scratch resistant plastic lens surface, as standard.

### Kodak Precise™ Progressive Lenses

**Kodak** Precise is our premium enhanced varifocal lens with a wide near zone for natural eye movement, a smooth transitional zone and clear distance zone. **Kodak** Precise Lenses offer clear natural vision at all distances, scientifically developed using a revolutionary new mathematical process called Vision First Design. Whichever way you look and at whatever distance, everything is sharp and with minimal distortion or blurring, similar to a single vision lens.

**Kodak** Precise Progressive Lenses are great for driving, reading, watching television, working on the computer - anything in fact where fast, clear natural vision is required.

Nearly all **Kodak** Precise Lenses are supplied with RLX Plus®, the world's most scratch resistant plastic lens surface, as standard.

### Kodak Unique Progressive Lenses

Everyone is different. Our face shapes vary from round to square and all other variables in between. Our head sizes vary, our skin tones are different and hence there are thousands of frames to choose from, so that we can find the perfect one for each of us as an individual. **Kodak** Unique is a lens that offers that level of individuality.

The **Kodak** Unique Lens is an innovative varifocal lens designed to be as individual as you are. Every pair of **Kodak** Unique Lenses is designed specifically for you, taking into

account your prescription and frame choice resulting in a pair of spectacle lenses closely tailored to your visual requirements. Unlike traditional varifocal lenses, the individual design of the **Kodak** Unique Lens means that a wider range of frames are open to you including many fashionable shallower frames. And it has been scientifically developed to minimise blurring or the 'swim' sensation associated with traditional progressive lenses.

**Kodak** Unique lenses are designed using advanced Vision First Technology. This and our state of the art manufacturing processes mean that your **Kodak** Unique lenses meet exacting standards of vision, individual to you.

### NIKON Progressive addition lenses availability

See series is Nikon's top-of-the-line range of lenses. SeeSeries is for eyeglass wearers who wish to enjoy the ultimate in visual performance.

SeeMax is available for both single vision and progressive lens wearers. It is a completely customized double aspheric lens offering the most superior visual performance and appearance possible. To permit visual quality upgrading in all areas of vision, Nikon's Aberration Filter Surface is applied to counter any significant aberrations and distortion that come with a wearer's prescription.

SeeStyle is the latest addition to the SeeSeries family of products. SeeStyle has been developed especially to accommodate the current trend in high-base frames. It is a new design that enables high-base lenses and high-base frames to match perfectly, affording superb optical performance and cosmetic appeal. To achieve this visual performance optimization, Nikon incorporates its Optical Correction System (OCS) into SeeStyle to overcome the visual discomfort that is frequently encountered with high-base frames.

SeeCoat is Nikon's most premium coating. Using clear protection technology, SeeCoat provides ultimate clarity, making the lens much more transparent. SeeCoat also contains a repellent coating layer resistant to fingerprints and dirt, which keeps lenses clean twice as long as conventional super-hydrophobic coatings. In addition, SeeCoat offers the world's highest level of scratch resistance, which makes it two times more resistant to abrasion than a conventionally-coated lens.

### Seiko Progressive lens availability

SS P1 Series (Custom made type)  
P-1 SY Series  
P-1 G Series  
P-1 W Series

### HOYA progressive lens availability

Hoyalux iD  
Hoyalux iD MyStyle



## Transition Coated progressive addition lenses (FAQ's)

### **Why can I no longer focus on objects up close?**

Vision changes occur naturally as you age. When you reach your 40's, focusing at close range becomes difficult. This is a natural part of the aging process caused by a gradual hardening of the eye's crystalline lens, reducing its ability to change shape and focus. This condition is called Presbyopia.

### **What are progressive addition lenses and how do they differ from ordinary bifocals?**

Progressive addition lenses feature a continuous, clear field of vision from distance, through the intermediate ranges, to near without the use of annoying bifocal lines.

Bifocal lenses use very old technology. Abrupt and awkward changes between distance and near prescriptions are separated by annoying lines that can be frustrating to wear.

### **Why are Varilux lenses better than other progressive lenses?**

Varilux has conducted extensive clinical research, the result of which has produced a lens offering all presbyopes naturally comfortable vision never before achieved with a progressive lens. With over 100 million pairs sold, the unique triple-patented\* varilux lens is the brand most prescribed by eye care professionals and trusted by millions of satisfied wearers.

### **Who would be a candidate for Varilux progressive lenses?**

Anyone who has presbyopia, including current bifocal and trifocal wearers .

### **Can I expect to have an adaptation period with my Varilux lenses?**

Most individuals will learn to use their Varilux lenses right away. Others may take a bit longer to become completely comfortable. We at American Vision Care should be made aware of any extended adaptation period. A "fine-tuning" frame adjustment may be all that is necessary to assist in this initial period.

### **I have a strong prescription. How can I reduce the thickness and weight of my eyeglass lenses?**

Varilux lenses are available in a variety of different materials that will reduce both thickness and weight. Smaller frames also reduce lens thickness and weight. We at American Vision Care will consider prescription, frame size, and your individual lifestyle as factors in helping you decide which lens material will be best for you.

**Do Varilux progressive lenses come with scratch coating and ultraviolet protection?**

Yes

**How do I know that I have received genuine Varilux progressive lenses?**

You should receive a Varilux Wearers Guide from American Vision Care when your eyeglasses are dispensed to you. A Certificate of Authenticity guaranteeing you've received genuine Varilux lenses is attached to the front cover of the Wearers Guide.

**What is the warranty on Varilux lenses?**

Warranties on Varilux lenses may vary among eye care professionals. It is important to understand warranties available to you from the eye care professional you plan to purchase your lenses from. We at American Vision Care warranty Varilux and all progressive lenses for a full 60 days, to insure you will adjust to them.

**How can I obtain more information about Varilux Lenses?**

In addition to the information provided here, and elsewhere on this web site, American Vision Care is an excellent source of information about Varilux lenses and their specific applications to your individual prescription and lifestyle.

**How do Transitions work?**

Transitions Lenses contain special photochromic dyes that cause the lens to darken when exposed to ultra-violet rays from sunlight. When the ultra-violet light diminishes, the lens becomes clear again. This photochromic effect comforts the eye and makes Transitions Lenses the right lenses in any light.

**Are Transitions Lenses an alternative to clear lenses?**

Yes, because Transitions Lenses are the ideal all-around lenses. While they are clear indoors, they adapt to any light condition when exposed to sunlight outdoors. In addition, they provide automatic UV protection, blocking 100% UVA and UVB rays. Transitions Lenses are also available for all types of prescriptions, from single vision to progressives, and are compatible with all frame styles.

**How fast do Transitions Lenses work?**

Transitions Lenses begin to darken the moment they are exposed to UV light. The degree of darkening is a function of the intensity of UV light. The instant UV rays are removed,

the lenses begin to clear. In short, this means that Transitions Lenses quickly darken when you go outdoors, and quickly lighten when you return indoors. Most users report that this transition happens so smoothly, they are unaware of the change. They are aware, however, of their continued visual comfort.

### **Why is UV protection important?**

Most of us make a point of taking all the necessary precautions to protect our skin against ultraviolet rays. But, although we are rarely aware of it, our eyes also demand protection against UV damage. We are all exposed to harmful UV rays, and, over the long term, these rays significantly contribute to the deterioration of our eyesight. We at American Vision Care therefore now recommend that people protect their eyes against UV exposure at all times. Transitions Lenses provide 100% protection against UV rays.

### **Can I use Transitions Lenses like sunglasses?**

In most outdoor conditions, Transitions Lenses are perfect for reducing the sun's glare. They actually lighten and darken as the sun goes in and out of the clouds. However, since extremely high temperatures tend to affect lens performance, Transitions Lenses may not always get as dark as sunglasses. Ask your eyecare specialist for advice.

### **Are Transitions Lenses suitable for driving?**

In a car, Transitions Lenses darken much less than they do outside because the windshield blocks most of the UV rays that cause the lenses to darken. Ask us at American Vision Care if you need a pair of prescription sunglasses for driving.

### **Are Transitions Lenses available in different lens designs?**

Transitions Lenses are available in a wide range of lens designs including Single-Vision, Bifocals and Progressives. They are also currently available in two different colors: gray and brown.

### **Can Transitions be worn while working on a computer?**

Transitions Lenses are perfect for this type of activity. In any situation where you'd wear regular lenses, Transitions can be worn to help you see more comfortably whatever you're doing.

## Frame selection for advance design Progressive Addition Lenses

The choice of eyeglass frames can make a difference to the ease and speed with which you adapt to progressive lenses. While getting used to the varying levels of vision now available to you through the progressive lenses, you will not need the distraction of ill-fitting eyeglass frames. This is especially important if the progressive lenses are made to a strong lens prescription, as anything other than a mild lens prescription may result in some temporary blurred vision or even dizziness until you adapt to the progressive lenses. Choosing well-fitting eyeglass frames will make it easy to keep your glasses firmly in place while you and your eyes are going through this period of adjustment.

An optometrist should be able to help with the selection of the eyeglass frames. While it is natural for you to want frames that suit your face, coloring and general style, keep in mind that fit is even more important. Eyeglass frames for progressive lenses once had to be fairly large to provide enough space for the various areas of focus, sometimes with the resultant cutting off of the close-up vision section at the bottom of the progressive lenses. Since then, however, the manufacture of progressive lenses has developed to the point where they are now more compact and can be used in smaller eyeglass frames if the wearer prefers this style.

## Caring of Progressive addition lenses

As well as providing with improved vision, progressive lenses are easy to care for. Just a little maintenance will keep them clean and ensure they give you the best service possible. A few common sense precautions will also protect progressive lenses from scratches that could harm their surfaces.

As well as using such products as lens spray, gel and towelettes, an easy way to clean progressive lenses occasionally is to dip your glasses in warm, soapy water. Just be careful not to use a greasy type of soap as it may result in a residue on the progressive lenses. Rinse the progressive lenses under running water, then use a lens cloth to pat them dry carefully. When using other products, try not to use any containing acetone or ammonia, because some progressive lenses may be sensitive to these chemicals.

One can also protect your progressive lenses by making sure you always put your glasses down so that they are resting on their arm pieces rather than on the lenses. Keep your glasses in their case when you are not using them and make sure the eyeglass case is the right size for the glasses, protecting them without putting unnecessary pressure on them. Be careful not to put your glasses too close to objects that are extremely hot or cold,

because extremes of temperature variations may damage both the frames and the progressive lenses.

## Progressive lens complaints

As well as providing you with improved vision, progressive lenses are easy to care for. Just a little maintenance will keep them clean and ensure they give you the best service possible. A few common sense precautions will also protect progressive lenses from scratches that could harm their surfaces.

As well as using such products as lens spray, gel and towelettes, an easy way to clean progressive lenses occasionally is to dip your glasses in warm, soapy water. Just be careful not to use a greasy type of soap as it may result in a residue on the progressive lenses. Rinse the progressive lenses under running water, then use a lens cloth to pat them dry carefully. When using other products, try not to use any containing acetone or ammonia, because some progressive lenses may be sensitive to these chemicals.

You can also protect your progressive lenses by making sure you always put your glasses down so that they are resting on their arm pieces rather than on the lenses. Keep your glasses in their case when you are not using them and make sure the eyeglass case is the right size for the glasses, protecting them without putting unnecessary pressure on them. Be careful not to put your glasses too close to objects that are extremely hot or cold, because extremes of temperature variations may damage both the frames and the progressive lenses.

## Fitting Tips

### Adjust the Frame

- Adjust the frame on the patient for maximum comfort and accuracy before taking any measurements.
- Set the vertex distance between 12 and 14mm.
- Set the pantoscopic tilt angle between 10 and 12 degrees.
- Frame should have positive facial wrap.

Always better to take monocular PD to ensure exact centering of the eye behind the lens. Ask consumer to look straight at optometrist's right eye when marking on left eye and vice versa.

At the time when taking near PD ask patient to look at optometrist's nose bridge portion and mark the PD.

### Take the Fitting Height Measurement

- Allow for frame slippage and avoid parallax error.
- Take a monocular height measurement by marking each lens at center pupil using a felt tip pen
- Draw a horizontal line on each lens and double-check to make sure that the lines are crossing the center of each pupil.
- Measure the fitting height from the deepest point of the lens to the pupil center.

### Check Lens Cut-Out

- Mark the patient's fitting height and distance PD on the sample lens, creating a cross
- Place the lens cross over layout-chart cross to verify that the lens will fit into the frame
- If the lens does not fit, choose another, more suitable frame that will accommodate the lens.

### Confirm Measurements and Prescription

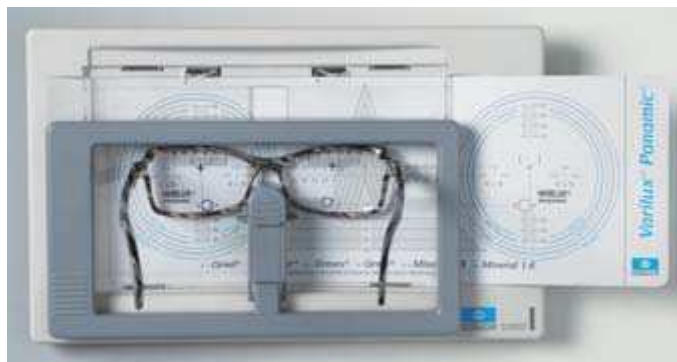
- Using the centering chart, center the frame over the inverted "V".
- Confirm the monocular PD and height
- Confirm lens prescription on lensometer

### Confirm Fit on the Patient

- With lenses marked or using decals, verify that the fitting cross is at pupil center.
- Adjust the frame to raise or lower the fit, if necessary

### Ditest

For optimal fitting of progressive lenses in the frames! Easy to use and precise!



## Troubleshooting

|                                                                  |                                                                               |
|------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Patient lifts head or glasses to read                            | Fitting too low or add too weak. Adjust frame or remake lenses                |
| Patient lowers head or glasses to look in the Distance.          | The lenses are fitting too high. Adjust frame or remake lenses.               |
| Patient has narrow reading area                                  | Add pantoscopic tilt and fit closer if possible<br>check fitting height       |
| Patient moves reading material off to the side for better focus. | Submit correct PD measurement and have lenses remade for optimum performance. |
| Peripheral vision blurs and moves                                | Increase facial wrap of frame                                                 |

## Conclusion

### Progressive Addition lenses for all

Optometry's core mission is to provide the best possible vision to our consumers. We do this in a number of ways, from tending to patients' ocular health to providing refractive correction. Good vision is also the bottom line for patients, particularly presbyopes.

Dispensing state-of-the-art eyewear affords the practice a golden opportunity to touch consumers' lives in meaningful and memorable ways. The eyewear we dispense serves as a daily reminder of the consumer's experience with us. If that experience was pleasant and the products we've dispensed enhance the patient's quality of life, we have not only done our job, we have opened the door to an ongoing relationship with that consumer and possibly to her network of family and friends.

In essence, optometry is about one thing: giving each patient the best possible view of the world. Since spectacle lens wearers comprise the majority of the patient roster, it only makes sense for the practice to have the most advanced tools at its disposal. For most presbyopes, general-purpose progressive lenses are that tool.

## REFERENCES

Clinical Optics Troy E. Fennin (OD) Theodore Grosvenor (OD, Ph.D.)

[www.ussafety.com](http://www.ussafety.com)

[www.googlewikipedia.com/progressivelenses](http://www.googlewikipedia.com/progressivelenses)

[www.zeiss.ru](http://www.zeiss.ru)

[www.wattonstreetoptical.com.au](http://www.wattonstreetoptical.com.au)

[www.earthvision.net/vision/13178progressivelenschoices](http://www.earthvision.net/vision/13178progressivelenschoices)

[www.drcarlryan.com/lenses.html](http://www.drcarlryan.com/lenses.html)

[www.essilor.co.uk/newsinfo/insight.pdf/insight06a.pdf](http://www.essilor.co.uk/newsinfo/insight.pdf/insight06a.pdf)

[www.americanvisioncare.com](http://www.americanvisioncare.com)

[www.seiko\\_opt.co.jp/i/products/lens\\_so2.html](http://www.seiko_opt.co.jp/i/products/lens_so2.html)

[www.hoyaluxid.com](http://www.hoyaluxid.com)

[www.cherryoptical.com](http://www.cherryoptical.com)

[www.articlesdirectorym.co.uk.solve\\_all\\_your\\_medical\\_needs\\_progressive\\_lenses](http://www.articlesdirectorym.co.uk.solve_all_your_medical_needs_progressive_lenses)

[www.solaone.com](http://www.solaone.com)

[www.variluxmd.com/variluxphysio.html](http://www.variluxmd.com/variluxphysio.html)

[www.americanoptometricassociation.com/aviationvision.html](http://www.americanoptometricassociation.com/aviationvision.html)

[www.opticare.com.au](http://www.opticare.com.au)

[www.variluxphysiosa.com](http://www.variluxphysiosa.com)

[www.variluxipseo.com](http://www.variluxipseo.com)

[www.varilux-university.org/E\\_LEARNING/E08\\_gav/Guide\\_EN.pdf](http://www.varilux-university.org/E_LEARNING/E08_gav/Guide_EN.pdf)

[www.nikon.com/products/ophthalmiclenses/progressive\\_lenses.html](http://www.nikon.com/products/ophthalmiclenses/progressive_lenses.html)

[www.kodak.com/products/ophthalmiclenses/progressive\\_lenses.html](http://www.kodak.com/products/ophthalmiclenses/progressive_lenses.html)